



Service Portfolio Measurement — A Framework for Evaluating the Financial Consequences of Out-tasking Decisions

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ABSTRACT

Topical developments in software-engineering facilitate the establishment of new design patterns for information systems. In Service-Oriented Architectures (SOA), processes of an information system can be extracted and “out-tasked” to service providers. KEEN/MCDONALD highlight the changes that are brought about by such an architecture with their statement „Out-tasking [...] breaks a company into a portfolio of process-centered operations rather than interlocking departments or functions.” [30]. Examples of technologies that have been developed for this purpose are COM+, CORBA und RMI [55]. With the initiative of Enterprise Application Integration (EAI), web-services turn out in practice to enable a widely spread realization of SOA.

With these technological achievements, new management tasks are arising in information systems science. As information systems are increasingly interlinked with other systems by various service providers, it is important to choose the appropriate composition of a corporate service portfolio. For this purpose, the long-term economic consequences of out-tasking decisions have to be taken into account. In this paper, we suggest a methodological framework for efficiency calculations that intends to suit for a proper evaluation of these consequences.

Due to the long-term consequences of information systems design, methods of capital budgeting are applied in the framework. Using Financial Plans (VOFI) [18], all payments driven by a decision can be taken into account, including various conditions for funding and loaning as well as taxes. We apply this method by analyzing typical in- and out-payments driven by out-tasking decisions that have to be taken into account throughout the life-cycle of service-oriented information systems. For evaluation purposes, efficiency measures need to be calculated. These measures indicate which composition of the service portfolio is most profitable in a certain corporate situation.

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In this paper, we demonstrate how to calculate the Total Cost of Ownership (TCO) as well as the Return on Investment (ROI) of a Service Portfolio on the basis of capital budgeting. Finally, perspectives are discussed for applying and extending the framework for Service Portfolio Measurement.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management---Cost estimation;
D.2.9 [Software Engineering]: Management---Life cycle; K.6.3 [Management of Computing and Information Systems]: Software Management---Software selection; K.6.4 [Management of Computing and Information Systems]: System Management---Centralization/decentralization.

General Terms

Management, Measurement, Economics, Standardization.

Keywords

Service-oriented Architectures, Service-oriented Business Applications, Portfolio Management, Portfolio Measurement, IT-Controlling, Total Cost of Ownership, Return on Investment.

1. Towards a Service Portfolio Measurement

1.1 Outsourcing

On today's markets, enterprises are increasingly forced to act efficiently and flexibly. To do so, there is a distinct trend for enterprises nowadays to concentrate in their business on core competences and strategic competitive advantages [[46] p.11]. Doing so, information systems play an important role as they enable new ways for doing business. A rather promising approach is to link companies together in networks by inter-organisational systems in a way that each company concentrates on a special field. For the single company, this means to source out fields of work by engaging another company to do the work. This concept of the so-called outsourcing is defined as follows: „the significant contribution by external vendors in the physical and/or human resources associated with the entire or specific components of the IT infrastructure in the user organization“[40]. In order to realise

an outsourcing strategy, an appropriate model for the coordination between the partners involved has to be found (cf. Fig. 1).

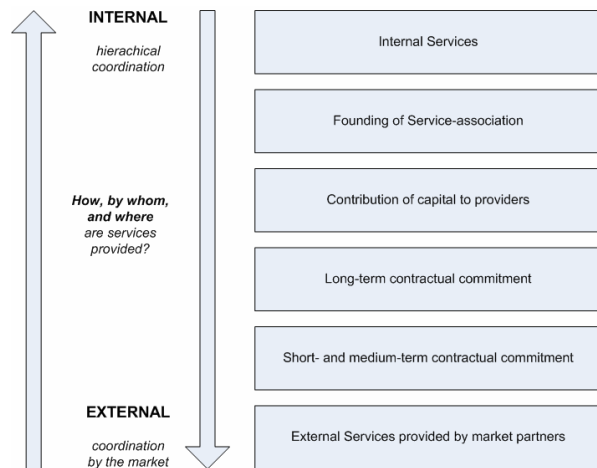


Fig. 1: Types of Coordination for Outsourcing [44]

Yet, the realisation of outsourcing also comes along with serious problems that have been increasingly pointed out in research by now [36]. Apart from problems concerning data protection, security, and controlling, also the danger of being dependent on the partners providing the outsourced work are reported. Service-oriented architectures of information systems seem to offer solutions for these problems.

1.2 Service-Oriented Architectures

With service-oriented architectures, information systems consisting of various components are intended to be built by showing both a high cohesion as well as a low linking [4]. In the past, technologies like COM+, CORBA and RMI have been developed for these purposes [[55], p. 18]. Nowadays, web-services turn out in practice to enable a widely spread realisation of SOA [16]. A reason for this might be the increasing need of integration technologies which arises in the course of Enterprise Application Integration (EAI). Further support might come from the work of standardisation initiatives.

There are many software solutions offered on the market which are already based on SOA. This fact can be regarded as a reaction to currents trend towards this new architecture. Examples are „Sonic ESB“ by Sonic Software [9], „mySAP Business Suite“ by SAP [50], „e-Business on demand“ by IBM [25] and the „Application Server“ by Oracle [43].

The design principles the implementation of SOA is based upon are given in Fig. 2. In addition to principles that are common for the design of information system architectures in general, special principals can be discovered that seem to be typical for the design of SOA.

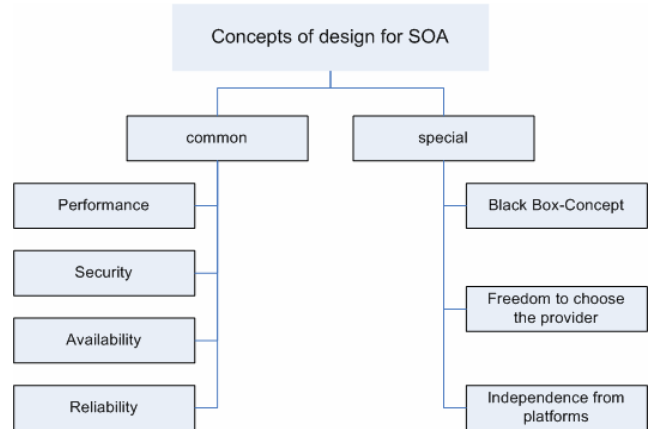


Fig. 2: Concepts of Design for SOA

The black-box-concept forms the basis for using SOA. Therefore the definition of input- and output-parameters of services is essential. Additionally, the interfaces used in order to integrate services should leave the free choice of service providers. Finally, it is important that services do not depend on special hard- or software platforms.

SOA gives the opportunity to concentrate on the core competences of a company without sourcing out complete fields of work, but by tasking out even specific sub-processes [48]. Doing so, it might be possible to use advantages of outsourcing by keeping the control over the entire process. This type of outsourcing is described by the term “out-tasking”.

1.3 Out-Tasking: Outsourcing Based on Service-Oriented Architectures

KEEN and McDONALD define out-tasking as follows: „Out-tasking demands a capability view of the business that breaks a company into a portfolio of process-centered operations rather than interlocking departments or functions. That portfolio is built on electronic collaboration the exchange of services across capabilities in the conduct of commerce. Out-tasking is enabled through application program interfaces (APIs) that electronically link to another company’s process capabilities. It is a term for using APIs to have another company handle a task on other behalf.“ [30].

Out-tasking transforms outsourcing [2], [3], [27] from an “all-or-nothing“-decision [7], [23], [10], [11], [52], [39] to the simple assignment of part processes (so called tasks) to specialised providers. [[32], p. 100]. The term “selective outsourcing” [[13] p.79, [38] p. 4, [42], [51], [1]] neatly illustrates the underlying concept.

Accordingly, the term out-tasking characterises on the one hand the outsourcing of single tasks to providers and on the other, the integration of electronic services of external providers [48]. In information systems supporting out-tasking some sub-processes are handled by the internal systems (in-tasked) and some processes are out-tasked to an external provider (out-tasked) (cf. Fig. 3).

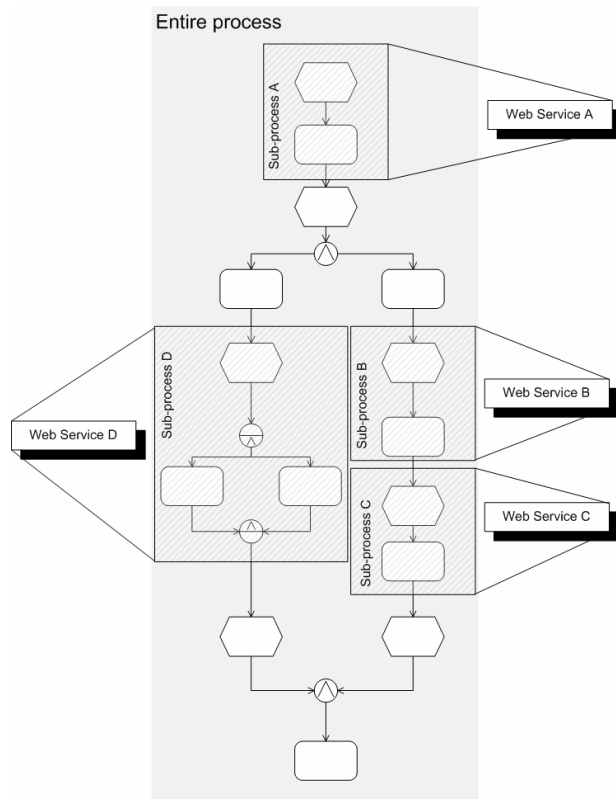


Fig. 3: Web Services as an Example for Out-Tasking of Sub-Processes

Problems arising from outsourcing can partially be avoided by information systems that are based on SOA. Due to the use of accepted standards, the technical dependence can be reduced, and by keeping the control of the process, the essential competences are kept in the company. The most benefit, however, might arise from the great variety of partners a company can choose for each specific service. As part-processes are sourced out, enterprises do not depend on few partners. On the contrary, they can build a portfolio out of several different service providers that fit to their contemporary demands.

With these technological achievements, new management tasks are arising in information systems science. As information systems are increasingly interlinked with other systems by various service providers, it is important to have an appropriate service portfolio management. On the whole, this means that a PDCA-Cycle has to be established on the appropriate composition of a corporate service portfolio. One major precondition for this is to develop appropriate methods for measuring the performance of such service portfolios [54] [22] [28]. This field should be referred to as the Service Portfolio Measurement.

The performance of service portfolios has to be analysed from a variety of perspectives. These perspectives can vary from business to business, examples being the strategic or financial consequences of a certain Service Portfolio.

For the development of methods for the Service Portfolio Management, different proceedings concerning the support of outsourcing-decisions can be taken into account.

A good overview is provided by KNOLMAYER, who considers proceedings about linear programming [comp. [49] p. 206, [31] p. 273, [53] [5]] under the following management approaches:

- “task specific cost comparisons [33]
- multi-task cost comparisons, including coordination costs [34]
- holistic cost-risk comparisons [56]
- portfolio models, partially based on transaction cost theory [45][12]
- flowcharts [6]
- checklists [26][8][29]
- cost-benefit analyses, partially based on pros and cons lists [35] and
- analytical hierarchy process models [47].” [comp. [32] p. 102].

Considering that measurement should be made out of various perspectives, the predominant task seems to be in finding out about what drives the efficiency of an out-tasking decision. As yet, the long-term monetary consequences associated with out-tasking decisions have not been taken into account. With this paper, we, therefore, suggest a framework for Service Portfolio Measurement from a financial perspective.

2. Evaluating the Financial Consequences of Out-tasking Decisions

2.1 Introducing a General Framework

In this chapter, we will present a general framework for efficiency analyses that can be used in order to evaluate the financial consequences of out-tasking decisions. The framework does not only give theoretical insight, but it may also serve as a tool for decision support in service portfolio management. Fig 4 gives the design of the framework with its methodological approach.

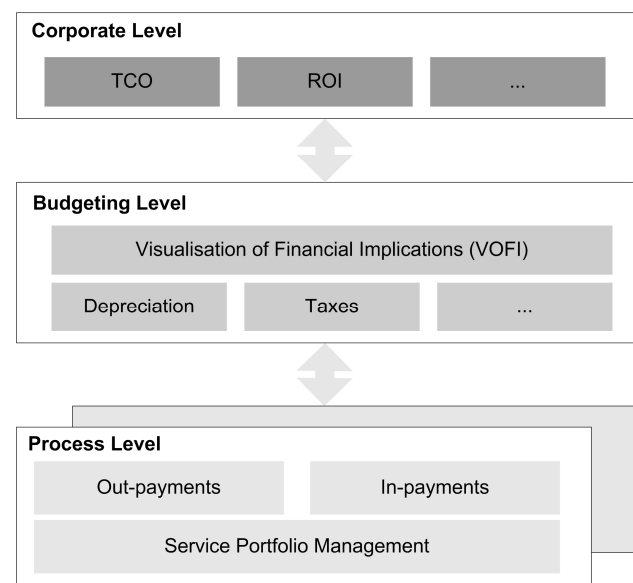


Fig. 4: Conceptual Framework for Evaluating the Financial Consequences of Out-Tasking Decisions

Due to the long-term consequences of decisions on the design pattern of an information system architecture, methods of capital budgeting are applied in the framework. On a macro perspective, three levels of evaluation can be distinguished: the process level, the budgeting level and the corporate level. The process level serves as a basis for the evaluation. Therefore, we analyse the payments and receivables brought about by the information system design by means of using a certain design pattern like SOA. On the budgeting level, additional parameters are taken into account that are relevant for judging the economic value created by series of payments. Relevant parameters are derived from specific conditions of funding and taxes that a company has to face. Finally, on the corporate level, the profitability of the information system design has to be compared to alternative investments available for the company. Measures like the Total Cost of Ownership (TCO) and the Return On Investment (ROI) help to consider relevant parameters for this purpose.

2.2 Consequences on the Process Level

The series of payments that are chargeable to a decision on the design of an information system architecture should be analysed from a life-cycle-perspective. Fig. 5 gives a general structure of the life-cycle of information systems services that might support identifying the relevant drivers of the efficiency of an out-tasking decision on the process level.

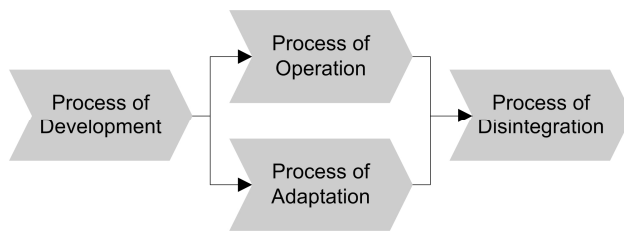


Fig. 5: Processes during the Life-Cycle of Information System Services

The processes will be introduced below by means of analysing characteristic payments and receivables to be considered in each process.

Process of Development

By analysing the series of payments brought about by an out-tasking decision, the individual context of the decision has to be taken into account. From a theoretical point of view, two typical context situations can be distinguished: projects for new- and redesigning of information systems.

Considering the new design of an information system, out-tasking already has an economical impact in requirements engineering. Whereas payments for the development of the out-tasked services can be saved, additional out-payments have to be taken into account for the design of a service-oriented architecture. These payments comprise for example those for the design of the interface as well as those for building the appropriate know how.

Another situation that might be relevant is the redesign of an information system with the purpose of supporting a service-oriented architecture. In such a project, the savings mentioned

above are not applied. As the tasks that are likely to be outsourced have been implemented already, the payments driven by them are no more relevant for the out-tasking decision. They are classified as so-called “sunk costs” [24]. In addition, further out-payments brought about by the work for redesigning have to be considered. As far as the functionality of the information system is not extended, these payments are totally to be charged to the out-tasking decision.

Phase of Operation

In total cost of ownership analyses, costs for the maintenance work on information systems are usually considered during operations [15] [41]. Against this background, it can be argued that by out-tasking services, the total payments for maintenance work are reduced by the equivalent for the work on these services. In contrast, additional payments have to be considered for the maintenance of the interfaces. Examples for this kind of maintenance work are adaptations to new versions of exchange formats on data and services.

Another important effect, however, is neglected in most of the analyses: the efficiency of the business processes that are enabled by the information system. In order to calculate these effects properly, GROB/VOM BROCKE suggest a method in which process models are used in order to identify relevant in- and out-payments brought about by a certain process design [17]. According to this concept, out-tasking creates the possibility to reduce the resources that are needed for running the information system. In order to calculate the amount of savings that are chargeable to a service, the values of all relevant resources as well as the rate in which they are used have to be considered.

On top of that, specific payments to the service provider have to be considered during the process of operation. Both, the amount of these payments as well as their distribution throughout the life-cycle clearly vary according to the model of pricing that has been agreed upon earlier.

Phase of Adaptation

During the runtime of an information-system, adaptations will have to be made to the system. These adaptations can be necessary in order to both, implement new services as well as to modify existing ones. Examples for drivers of such adaptations are technological innovations as well as changing demands.

Depending on the information systems architecture, different financial consequences of these adaptations have to be taken into account. In case that parts of the system are affected that are run by the company itself, out-payments for the development of changes to the system have to be charged. Relevant indicators are both the amount of man-months needed as well as the appropriate average cost rate to be calculated. In case out-tasked services are affected, these payments might be saved. However, it should be taken into account that the prices for the services provided might rise. Moreover, it is likely that changes might have to be made, that are not covered by service providers, so that services might have to be in-tasked by the company again.

As the need for changes is highly uncertain, methods of probability calculation should be applied here. One approach might be to estimate out-payments for adaptations in each period during the planning horizon, for example by using the scenario technique. To simplify these estimations, different types of

adaptations can be defined depending on their duration and the average amount of payment. These types of adaptations can be planned by estimating the probability and point in time they occur in.

Phase of Disintegration

Finally, it has to be analysed, whether specific financial consequences of out-tasking decisions can be foreseen in the phase of disintegration. A positive effect can result from the reduction of the resources that are needed in order to run the information system. Due to this effect, idle time costs [20] can be saved by disintegrating services. However, potential bounds by the contracts with service providers have to be taken into account. These bounds might both limit the flexibility of changing partners as well as bring about ongoing payments.

Series of Payments

All relevant parameters have to be aggregated in a series of payments. Fig. 6 sums up the typical payments driven by an out-tasking decision analysed above and shows how they are spread over the planning horizon.

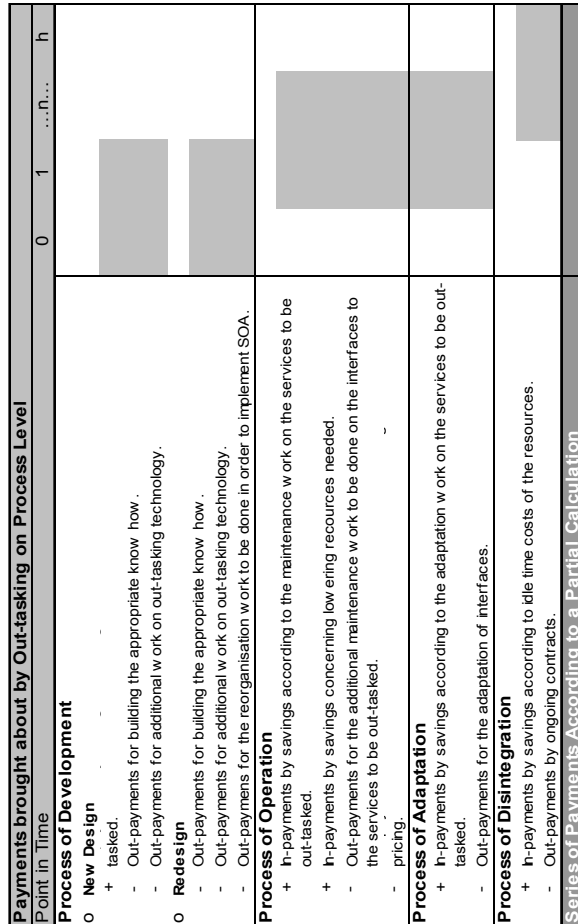


Fig. 6: Service Portfolio Measurement on the Process Level Identifying Payments

By means of aggregating the payments identified, essentially two alternatives can be distinguished: total or partial calculation. In order to make a total calculation, all payments chargeable to an information system with a certain service portfolio have to be accumulated. In order to choose the appropriate mix of services, all alternative portfolios will have to be calculated separately. This way, each portfolio will be evaluated in its own VOFI with its own efficiency measures. In practice, there are often mainly two alternatives: with or without out-sourcing. In this case, partial calculation can be made, in order to reduce the amount of data and operations needed. In partial calculation, only the different amounts in payment brought about by the two alternatives are aggregated in the series of payments. When calculating the difference between the alternatives by the pattern “with-case minus without-case”, a system design that is using out-tasking is profitable if the resulting measures are positive.

In both ways, the series of payments have to be aggregated over time regarding additional financial parameters that are relevant for the evaluation of economic consequences of out-tasking decisions.

2.3 Consequences on the Budgeting Level

On the budgeting level, the financial consequences are measured that arise from providing the payments needed during the life-cycle. For their analysis, the method of “visualisation of financial implications” (VOFI) is applied [18].

Using VOFI the financial consequences of long-term decisions are structured and calculated by means of spreadsheets that serve as a database for further analysis. Compared to formulas applied by conventional methods of capital budgeting (e. g. Present Value or Annuity of an Investment Project), VOFI mainly has the following advantages for evaluating the financial consequences of out-tasking decisions:

- **Transparency:** The description of the financial consequences of decision making by using spreadsheet increases help to show the various underlying impacts on the efficiency measures. By means of VOFI, all payments driven by a decision can be taken into account comprehensively, including various conditions for funding and loaning as well as taxes.
- **Adaptability:** VOFI serves as a reference model or long term decision making. Due to the explicit description of the financial consequences, it can easily be adapted to special decisions that have to be evaluated. On top of individual rates for loaning and funding, also dynamically changing conditions like tax rates can be calculated. Due to clearly defined interfaces for the in- and outflow of data, also extensions of the framework can be build.

Both the transparency and the adaptability finally might contribute to the actual use of efficiency calculations in service portfolio management in practice.

Customising VOFI specific parameters that are relevant in a certain capital situation (e.g. funding conditions) can be considered. Typical parameters that may serve as a reference are summed up in the VOFI given in fig. 7.

VOFI for Localisation Projects				
Point in Time	0	1	...n...	h
Series of Payments				
Internal Funds				
- Withdrawals				
+ Deposits				
Installment Loan				
+ Credit Intake				
- Redemption				
- Debitor Interest				
Annuity Loan				
+ Credit Intake				
- Redemption				
- Debitor Interest				
- Creditor Interest				
Loan in Current Account				
+ Credit Intake				
- Redemption				
- Debitor Interest				
Financial Investment				
- Reinvestment				
+ Disinvestment				
- Creditor Interest				
Tax Payments				
- Out-Payment				
+ In-Payment				
Net Funding	0	0	0	0
Balances				
on installment loan				
on annuity loan				
on current account				
on financial investment				

Fig. 7: Service Portfolio Measurement on the Budgeting Level Using VOFI

In order to consolidate the various influences on the effectiveness of the decision over time, a periodic update of the capital stock has to be calculated. Starting in period zero, each period has to be calculated in a way that there is a balance between in- and out-payments. The following example may illustrate the essential procedure. In the first period usually an out-payment has to be financed. If the internal funds available are insufficient, a loan has to be taken out. As usually various conditions for loaning can be agreed upon also a combination of various loans can be calculated in the VOFI. Correspondingly, multiple forms of funding can be included.

By calculating the adequate amount of loaning or funding, also tax payments have to be taken into account. In order to calculate tax payments, an auxiliary calculation has to be carried out. Depending on tax law, relevant parameters are for example individual depreciations that are chargeable to the investment as well as tax rates. In each period these periodical in- and out-payments have to be balanced. As a check-up, the net funding value, which is defined as the accounting balance of all out- and in-payments, should be zero.

On the basis of this flow figures mentioned above, the capital stock can be updated periodically. For this purpose, the balances of all loans and funds have to be recorded. The accounting balance of both of them finally results in the net balance of the total investment. By this algorithm the value of an investment in a service-oriented architecture can be monitored during the whole life-cycle of the information system simply by observing the net balance in each relevant period. For special interest investigations additional efficiency measures can be calculated on the basis of the VOFI.

2.4 Consequences on the Corporate Level

In order to make proper decisions, also opportunity costs have to be taken into account. This means to compare the financial results

calculated with VOFI to alternative investment available for a company. To do so, a wide range of efficiency measures can be applied from accounting science. Taking the Total Cost of Ownership (TCO) and the Return on Investment (ROI), relevant parameters for evaluating the efficiency of investments in the design of the information system architecture can be shown.

Total Cost of Ownership (TCO)

In case of consolidation of the payments was done on the basis of a total calculation, the total costs of ownership brought about by a certain information system architecture can be calculated. The measure aims to sum up all relevant costs chargeable to an information system throughout its life-cycle [15]. By using VOFI, the total cost of ownership can be calculated by summing up the payments of each row on the spreadsheet and balancing them like in fig 8.

Total Cost of Ownership	Value
Revenues	
+ Irregular Revenues	
- Regular Expenditure	
- Depreciation	
= Total Monetary Profit I (before Interest and Revenues by Complementary Investments)	
+ Revenues by Complementary Investments	
- Interest Expense	
= Total Monetary Profit II (before Taxes on Income)	
+ Taxes on Income	
= Total Monetary Profit III (after Taxes on Income)	
- Calculatory Interest	
+ Adjustment of Tax Payments	
= Total Calculatory Profit	
= Total Cost of Ownership	

Fig.8: Service Portfolio Measurement on the Corporate Level Using TCO

Calculating the TCO, the monetary profit of the investment can be calculated gradually. Starting from the regular revenues, first irregular revenues, then regular expenditure and finally the depreciation is charged resulting in the Total Monetary Profit (I). Considering revenues by complementary investments made during the life-cycle as well as interest expenses, the Total Monetary Profit (II) can be calculated. By charging tax payments on income, the Total Monetary Profit (III) can be reported. While these measures all show the monetary profit realised by the investment, also alternative allocations of the capital have to be taken into account for consistent decision making. For this purpose, calculatory interest has to be charged quantifying the profit that is assumed to be made by the so-called opportunity. This Total Calculatory Profit finally represents the Total Costs of Ownership calculated on the basis of VOFI. Usually the resulting value will be negative – apart from the revenue of potential liquidation of the system – as only out-payments are charged. For reporting the Calculatory Profit as the Total Costs of Ownership, the value should consequently be transformed into a positive value.

Using the Total Cost of Ownership, out-tasking is profitable, if it serves to reduce the TCO. Moreover, the measure supports designing the appropriate corporate service portfolio. Accordingly, the portfolio should be designed with the aim of minimising the TCO.

Return on Investment (ROI)

The efficiency of out-tasking decisions can also be evaluated by a measurement of profitability. A widespread measure of this kind in practice is the Return on Investment (ROI). With this measure, a ratio is calculated that sets the total profit in relation to the stock of capital provided for the investment. Consequently, the ROI indicates the interest payment made by an investment. Whereas the ROI seems to be suitable to compare investments of different kinds, it is highly criticised in research [17]. One of the reasons for criticism lies in the fact that it is a static measure and therefore inappropriate to support long-termed decisions [18]. This point can be considered by calculating the ROI on the basis of VOFI. In doing so, the ROI is not only a dynamic measure, but it also considers various conditions of loaning and funding as well as taxes. Fig. 9 shows the definition of this ROI.

Target	Symbols
$g_{GK} = \sqrt[n]{\frac{EW^M + FK_0 + Z^S}{EK + FK_0}} - 1 \quad \text{für } EW^M + FK_0 + Z^S \geq 0$	r_{GK} ROI, dynamic k Capital Costs Ratio EW^M Net Balance in t=h EK Internal Funds FK_0 External Funds Z^S Total Creditor Interest
Benchmark	
$k = \sqrt[n]{\frac{EK + Z^O + FK_0 + Z^S}{EK + FK_0}} - 1$	
Recommendation	
Out-tasking is financially profitable, if $r_{GK} > k$	

Fig. 9: Service Portfolio Measurement on the Corporate Level Using ROI

In particular, the ROI can be used for decision support on the design of the information system architecture, providing the series of payments are partially calculated. For example ROI is appropriate for evaluating whether the payments necessary for the migration to a service-oriented architecture are justified by future savings.

3. Conclusion

With this paper we suggested a general framework for the evaluation of financial consequences of out-tasking decisions. By applying this framework, we identified typical drivers for the profitability of these decisions. As a basis, relevant types of in- and out-payments brought about by out-tasking were analysed throughout the life-cycle of an information system. As the decision has long-term consequences additional financial parameters were examined. An important influence on the profitability comes from the capital stock of the company as well as from the tax law. Finally, it was shown that principles taught by decision making theory have to be taken into account in order to choose the most profitable design of service portfolio for a company. Essentially, this means to take a calculatory perspective on the investment finally. For this reason investment in the information system design has to be compared to alternative investments available for the company. With these findings, it can be argued that out-tasking decisions have to be judged by efficiency measures, such as the Total Cost of Ownership as well as the Return on Investment. The Framework provided in this paper therefore may serve as a

reference model for decision support in service portfolio management.

Further research should concentrate on the extensions of the framework as well as on its empirical use in practice. Against the setting of our evaluation, it seems to suggest itself to further include methods of probability calculations for example. Especially the risk analysis seems to be a promising approach for this purpose [21]. Another natural extension can be seen in including additional efficiency measures.

Concerning empirical studies, the design of present service portfolios should be measured regarding their efficiency. On this basis, individual advice can be given in order to optimise service portfolios. In addition to checking portfolios, support for future out-tasking decisions might be provided by the methodology described with this paper.

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